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Linear electrostatic actuators with Moiré-effect optical proprioceptive sensing and electroadhesive braking

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Abstract

Muscles in animals and actuation systems in advanced robots consist not of the actuation component alone; the motive, dissipative, and proprioceptive components exist in a complete set to achieve versatile and precise manipulation tasks. We present such a system as a linear electrostatic actuator package incorporated with sensing and braking components. Our modular actuator design is composed of these actuator films and a dielectric fluid, and we examine the performance of the proposed system both theoretically and experimentally. In addition, we introduce a mechanism of optical proprioceptive sensing utilizing the Moiré pattern innately generated on the actuator surface, which allows high-resolution reading of the position of the actuator without noise. The optical sensor is also capable of measuring the force exerted by the actuator. Lastly, we add an electroadhesive brake in the package in parallel with the actuator, introducing a method of mode switching that utilizes all three components and presenting control demonstrations with a robot arm. Our actuation system is compact and flexible and can be easily integrated with various robotic applications.

Keywords

Electrostatic actuator, electroadhesive brake, Moiré effect, optical sensing, proprioceptive actuator, modular actuator

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1. Introduction

Robots with increased performances and utility require actuators with compact form factors and high force densities. Such characteristics are especially critical for mobile robots in spatially constrained and cluttered environments (Sofla et al., 2010) and wearable robots for physical human augmentation (Asbeck et al., 2014; Park et al., 2014; Kwon et al., 2019; Thalman and Artemiadis, 2020; Wirekoh et al., 2020) and haptic feedback (Chossat et al., 2019; Ham et al., 2023; Pacchierotti et al., 2017). Although most existing robotic systems are operated by electromagnetic actuators for practical advantages, such as high force density and short response time, these actuators inevitably require thick and high-density metal components, such as bulky coils and permanent magnets, to amplify the strength of magnetic field and the actuation force. Therefore, it is challenging to realize a form of an actuator that is both compact and flexible with a mechanism similar to that of electromagnetic actuation.

Electrostatic actuators have several distinct advantages over electromagnetic actuators. They create meaningful magnitudes of Coulomb force at small distances between electrodes, allowing for overall thin, micro-scale structure designs. A major application field of electrostatic actuators thus has been micro-electromechanical systems (MEMS) (Algamili et al., 2021; Muralt, 2000). Moreover, these actuators utilize polymer-based (Kornbluh et al., 1999; Shankar et al., 2007) and oil-based (Ge and Ludois, 2016; Niino et al., 1994; Rothemund et al., 2021) dielectric media with high relative permittivity for force amplification. Their non-metallic materials not only provide flexible and thin structures but also allow high specific force during actuation. Lastly, for the same amount of mechanical power generation, electrostatic actuators generally require a higher voltage but a lower electric current than electromagnetic actuators. The lower current creates the lower electromagnetic

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interference, which is useful in MR-compatible environments (Gassert et al., 2006; Rajendra et al., 2008).

Efforts have been made to realize large-scale displacement and force while taking advantage of the above features of the electrostatic actuators (Hosobata et al., 2015; Niino et al., 1993; Rothemund et al., 2021; Yamamoto et al., 1998). In particular, previous studies introduced linear film actuators that utilize electrostatic force by supplying alternating voltage inputs to repeatedly patterned micro-scale electrodes (Hosobata et al., 2015; Niino et al., 1993; Yamamoto et al., 1998). Those actuators with a large number of repeating electrode patterns create linear motions and forces that are applicable to large robotic applications beyond the typical scales of MEMS devices. However, research on such film actuators has focused on the electrode designs and the modeling of force, but for actuators evaluated in restrictive planar environments with rigid guide rails (Niino et al., 1993; Yamamoto et al., 1998). One of the strongest advantages of these film-type actuators is in its thin and flexible feature. Therefore, to fully utilize these characteristics, it is important that the actuator is designed and packaged as a complete module. As a module, the actuator becomes easily integratable with robotic systems, realizing a form that is both compact and flexible.

In addition to generating motion and force, there are other design considerations when developing the actuator module from practical perspectives. Biological muscle and actuation systems in advanced robots, for instance, own adequate position and force sensing components for proprioception (Ott et al., 2016; Tuthill and Azim, 2018). With their proprioceptive sensing capabilities, these systems implement closed-loop control of displacement and force (Wirekoh et al., 2019). Also, these existing actuation systems have braking and clutching abilities used to effectively deal with dissipative and holding motions (Brunello et al., 2007; Dickinson et al., 2000; Kwon et al., 2022; Plooij et al., 2015; Shafer and Kermani, 2010). By adding appropriate sensing and braking elements that match the form factor of the electrostatic actuator, the actuator module can be utilized in a wide field of robotics.

We propose a fully-packaged linear electrostatic actuation system (Figure 1). The system consists of a flexible electrostatic actuator, an optical sensor for closed-loop control, and an electrostatic brake for holding motions. The package design of our actuator module increases the practicality of existing electrostatic film actuators for applications in robotics with multiple functionalities of actuation, position and force sensing, and clutching, while maintaining a thin and compact form.

This paper describes the concept of the actuation system and explores each component in the system with theoretical and experimental evaluations. Section 2 discusses the main design of the electrostatic actuator. After derivation of the

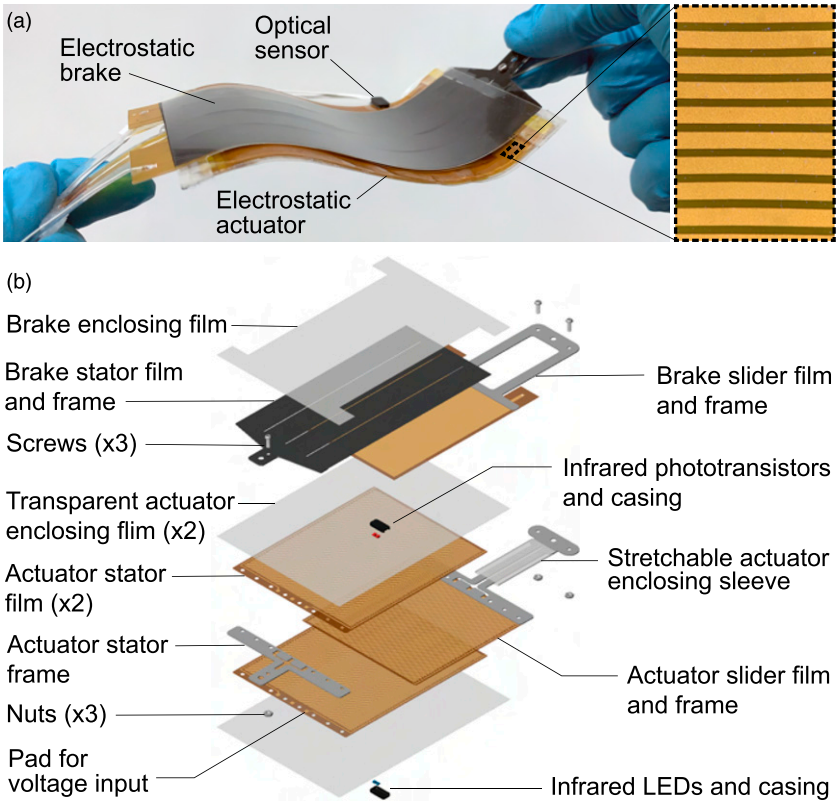


Figure 1. (a) The proposed actuator package consists of an electrostatic actuator, an electrostatic brake, and a sensor. The package is thin and flexible. The enlarged figure shows the micro-scale electrodes in actuator films for three-phase synchronous actuation. (b) Exploded view of the actuator package and its constituent components.

electrostatic force output models, the electrode designs based on the models are decided. This section also introduces a sealed actuator package enclosing a dielectric fluid and evaluates the performance of the actuator. Section 3 introduces the principle of the optical sensing based on the Moiré patterns generated on the actuator surface. With the integrated sensor, high-resolution position and force sensing is achieved. Section 4 discusses integration of the electrostatic brake into the actuation system and demonstrates the mode change between actuation and braking. Closed-loop control with a single degree-of-freedom (DOF) manipulator shows the effectiveness of the combination of the three components. Section 5 finally concludes the work with discussions on the contributions and the future work.

2. Actuator

Coulomb's law states that there exists a force between two charged particles, also referred to as electrostatic force. Supplying or removing the charges by providing the difference in the electrical potential changes the force between the two electrodes. By patterning the electrodes in a periodic manner and controlling the voltage of the electrodes accordingly, continuous actuation is achieved which, in principle, is analogous to actuation of electromagnetic synchronous motors based on Lorentz forces (Toliyat and Kliman, 2018).

Higuchi and Yamamoto have presented detailed modeling and working prototypes of such synchronous electrostatic motors (Yamamoto et al., 2006). Two main parts, the slider and the stator, were designed with thin, long patterns of electrodes enclosed in polyimide layers to avoid a short circuit when assembled. By applying alternating high-voltage waveforms to the electrodes with different phase offsets, Coulomb forces between the electrodes produce a relative linear motion between the slider and stator such that it is synchronous to the apparent speed of the traveling electrical potential.

Our proposed actuator is founded on the same grounds. It uses the periodic pattern of the electrodes in the stator and slider configuration to provide a continuous synchronous linear motion by applying high-voltage sinusoidal waveforms with evenly distributed phases to the electrodes. In particular, the voltage waveforms are supplied as a single three-phase source to both the slider and the stator with opposite configurations, similar to the design in Niino et al. (1993). The speed of the traveling electrical potential is therefore twice that of the supplied waveforms.

Since synchronous actuators move at speeds matching the traveling electrical potential, their actuation force depends on the positional difference between the electrical potential and the actuator, also called the phase lag, instead of simply the actuator position alone. Typically, a synchronous actuator exhibits a force that is sinusoidal to this phase lag and reaches the maximum pull-out force at 90°.

The modeling of the output force and the electrode design of our electrostatic actuator are provided in the Appendix.

2.1. Compact form factor

Based on the actuator model, we decided the dimensions of the electrode for the stator and the slider of our prototype (Figure 2(a)). For both actuator films, the length L, the thickness t, and the pitch p of the electrode were 200 µm, 12 µm, and 300 µm, respectively, providing the electrode length ratio of $\lambda = L/p = .666$. The actuator films were fabricated using a conventional fabrication method for double-layered flexible printed circuit boards (fPCBs). All the electrodes were located on the same layer, and the other layer was used for providing via holes to electrically connect one of the three-phase electrodes. Overall, the thickness of the actuator films was 100 µm. The electrodes were made from copper, and the base and the cover layers were made of polyimide.

The width of the layer was 80 mm, and most of the layer width (74 mm) was covered by the electrode for both the stator and slider films. The shapes of the electrodes were, however, different in skewing for the two films (Figure 2(b) and (c)). While the slider film had a straight pattern, the stator electrodes had a zig-zag pattern with four consecutive V-shapes with a skew factor of $s_f = 1$, which is defined as the ratio of the skewing length along the x-axis to the electrode pitch p. In previous work, it has been shown that the number of the V-shapes affected the stability of the actuator against yawing, and that the actuator was stable when the number was larger than four (Yamamoto et al., 1998).

Our actuator was made of two stator films and a slider film in a sandwiched structure (Figure 2(d)). The sandwich structure has an advantage of amplifying the output force by utilizing both sides of the slider film. Since the electrodes were coated on one side of each film, all three films needed to be stacked in the same direction, facing downward, to maintain the same distances between electrodes of different

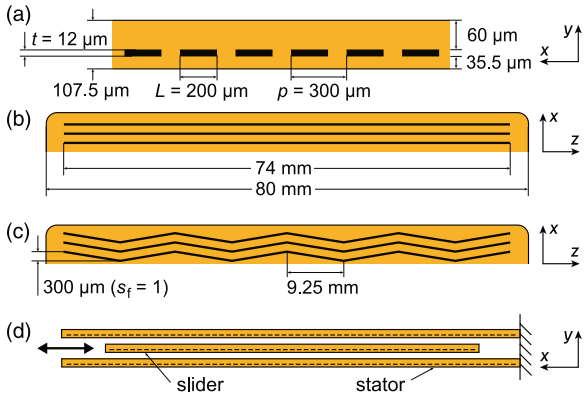


Figure 2. (a) Side view of the stator and slider films and their dimensions. (b) Front view of the slider film and its dimension. (c) Front view of the stator film and its dimension. (d) In our actuator, one slider film is superimposed between two stator films.

films, as shown in Figure 2(a). The lengths of the stator and the slider were 100 mm and 80 mm, respectively, and the difference of 20 mm provided the range of linear displacement that kept the same overlapping area. The weights of the stator and the slider were 1.4 *g* and 1.1 *g*, respectively, making the actuator extremely lightweight.

2.2. Sealing/packaging

The performance of electrostatic actuators has been improved using dielectric fluids (Ge and Ludois, 2016; Legrand et al., 2006; Niino et al., 1993). Such fluids provide higher dielectric constants than air, enabling the actuator to generate a higher force than that with an air gap at a given input voltage. The high dielectric strength of a dielectric fluid also prevents electrical breakdown between electrodes. As a result, actuators with a dielectric fluid not only provide high actuation forces due to an increased effective dielectric constant but also can be driven at high-voltage inputs, further improving the output force. Another advantage of using a fluid is formation of lubricant layer that reduces the stiction between the electrodes in spite of the viscosity of the fluid. While the viscosity and the adhesion of dielectric fluids degrade the actuator performance at high driving frequencies, they ensure that the stator and the slider remain in a stable contact even when electrical power is not supplied. Maintaining a stable contact between the films is critical to both consistent starts and continuous operation, especially for actuators that are thin and flexible. Other sealed electrostatic actuators that encase dielectric fluid, such as the HASEL artificial muscles (Rothemund et al., 2021; Acome et al., 2018), also utilize the incompressibility of the fluid to change the actuator shape and to generate the displacement, but the inherent requirement for a large volume of fluid results in a thick form factor; the dielectric fluid in our actuator whereas is used to more effectively drive the films and therefore only a small volume is required.

However, due to the difficulty in sealing, dielectric fluids have been widely used only as part of certain experiments rather than as an integrated component of an actuator module. Figure 3 shows a package of our flexible actuator containing a dielectric fluid sealed inside. Two stator films and a slider film were attached to their respective frames made of polyethylene terephthalate (PET) sheet (thickness: 0.5 mm) shaped by a laser cutter (Speedy 300 Flexx, Trotec). The input wires were soldered to the pads on the stator films and the slider film, and the wires were routed along the spaces designed at the center of the stator frame and the slider frame, respectively. The actuator films were then enclosed in a flexible bag made of two plastic sheets by heat sealing the edges. This flexible casing not only prevents the dielectric fluid from leaking but also allows the motion of the slider only in the *x*-axis direction, playing a role of a linear guide. The slider film is the only component that changes the overall shape of the actuator during actuation. We thus added a stretchable and flexible sleeve made of

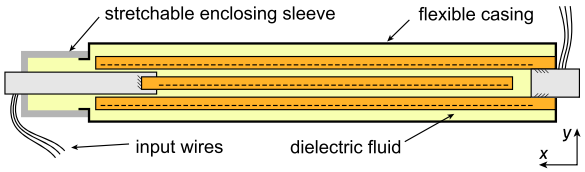


Figure 3. Stator and slider are enclosed in a flexible casing and immersed in a dielectric fluid. The slider side casing has a stretchable sleeve.

elastomer (Ecoflex 00-30, Smooth-On) to allow the linear motion of the slider. The cylindrical sleeve was fabricated with a wall thickness of 0.1 mm and has a natural length when the actuator is at the center of its range of motion. Due to its thin wall and low modulus, the sleeve created a low resisting force, less than 0.1 N, when the slider was located at both ends of the range of motion. A silicone adhesive (Sil-Poxy, Smooth-On) was used to glue the flexible casing and the stretchable sleeve together. After assembling all the components, 3 mL of dielectric oil was injected into the casing using a syringe with a hypodermic needle. We used off-the-shelf corn oil which showed the best performance in comparison to different types of oils, such as silicone oil, castor oil, and an industrial transformer oil (MICTRANS, Michang Oil), based on the preliminary experiments. The complete actuator prototype package was less than 1 mm thick and weighed 9.5 *g*.

2.3. High-voltage system

We operated our three-phase electrostatic actuator using a custom-built high-voltage power supply, shown in Figure 4(a), which had six high-voltage output channels that are programmable independently. By using three of these channels and controlling the amplitudes, the frequencies, and the phases of the input signals, we were able to control the actuation force, the actuation speed, and the actuation direction of our three-phase actuator, respectively. While typical transformers have rated ranges of the operation frequency for continuous voltage amplification, our custom-built power supply provides high-voltage outputs with a wide range of driving frequencies as well as high-voltage direct-current (DC) outputs.

Figure 4(b) shows the functional circuit diagram of one channel of the high-voltage power supply. First, a unipolar alternating current (AC) signal (i.e., a sinusoidal wave with a DC offset) is generated from a microcontroller and a digital-to-analog converter (LTC1655, Analog Devices). The signal is then fed into a power operational amplifier (PA75CD, Apex Microtechnology) in a unity gain buffer scheme. The power output from the amplifier goes through a DC to high-voltage converter (AH60P-5, XP Power) and then contains voltages in the order of magnitude capable of driving our electrostatic actuator. Two current-bleeding resistors are connected in series between the high-voltage output and the high-voltage return for fast discharging. These resistors are also used to monitor the high-voltage

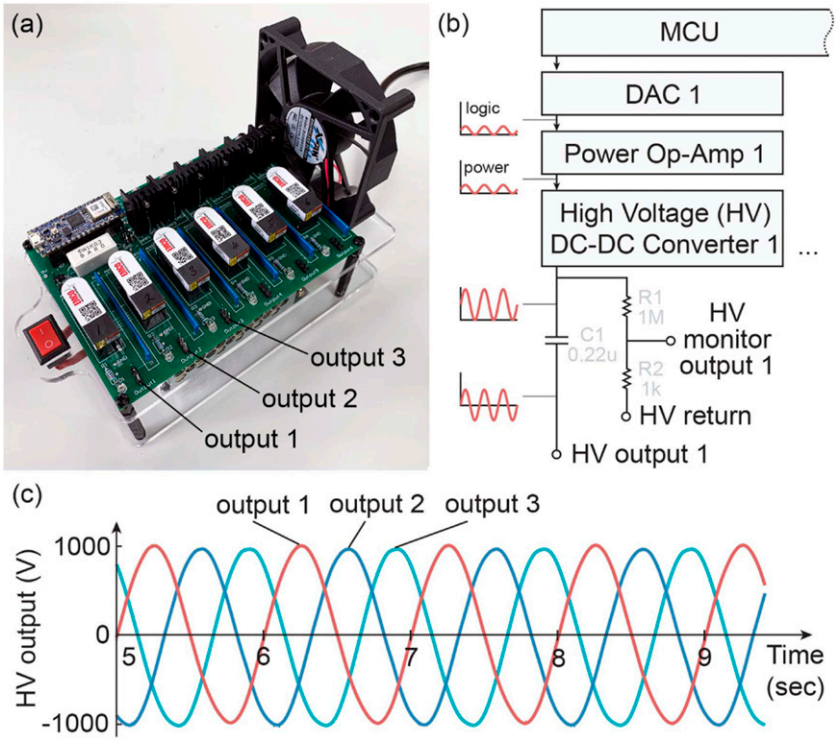


Figure 4. (a) Multi-channel programmable power supply. (b) Functional circuit diagram of one high-voltage output channel. There are a total of six output channels in the power supply. (c) Measured 3-phase voltage outputs of 1000 V at 1 Hz.

outputs in a voltage divider circuit. A capacitor connected at the output converts the unipolar high-voltage to be bipolar before finally being supplied to the actuator electrodes. All the high-voltage return terminals are connected together and grounded to earth through a 1 kΩ power resistor.

Figure 4(c) shows the three-phase voltage outputs with an amplitude of 1000 V at 1 Hz. The bipolar outputs were measured using an external voltage divider and an oscilloscope. Since the voltage output of the DC to high-voltage DC converter is slightly load dependent, we calibrated the voltage amplitude of each channel to help compensating the load by reading the monitoring signal while connecting and driving the actuator.

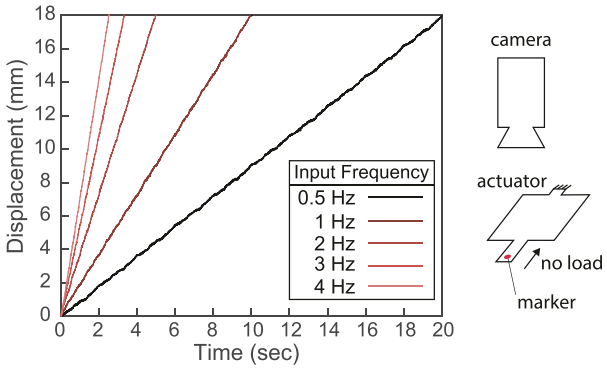


Figure 5. Actuator speeds at various input frequencies. The voltage amplitude was set to 1000 V. The average actuator speeds match with the synchronous speed, 6*pf*.

2.4. Experimental validation

We conducted two experiments to evaluate the basic performances of the actuator. First, we measured the actuation speed in a no-load condition. The actuator was fixed on a table, and a marker was attached to the end of the slider frame. Using a camera, we filmed and tracked the movement of the marker at a sampling rate of 240 Hz, calculating the marker position using a motion-graphics software tool (Blender, Blender Foundation). Figure 5 shows the actuator speeds under various driving frequencies. The average actuator speeds were 0.9, 1.8, 3.6, 5.4, and 7.2 mm/s at 0.5, 1, 2, 3, and 4 Hz, respectively; these follow the synchronous speed, 6*pf*, proportional to the electrode pitch p and the supplied AC frequency f (Yamamoto et al., 1998).

The actuation force was then measured using a force gauge (Series 3, Mark-10). Since the actuation force is not constant but changes depending on the actuator position, as depicted in Figure 22 in the Appendix, we connected a rubber band between the actuator and the force gauge to measure the steady-state peak forces when driving the actuator at fixed amplitudes and frequencies of the voltage input. As shown in Figure 6, the steady-state peak force was proportional to the square of the amplitude of the applied voltage. When 1400 V was applied at 0.5 Hz, the force output was 5.8 N, resulting in a large specific force of 610 N/kg. Although the electrostatic force was not related to the applied frequency in our model, the experimental results

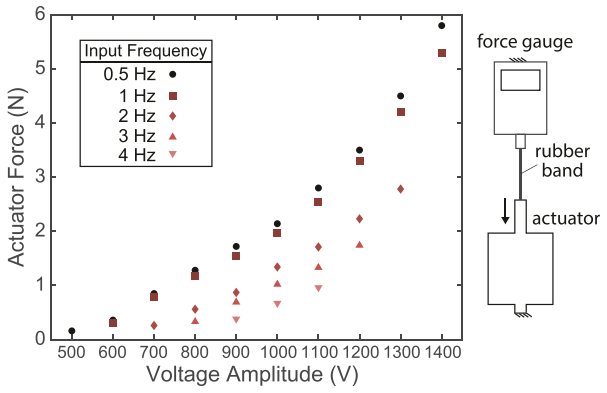


Figure 6. Measured actuator forces at various input voltage amplitudes and frequencies. The actuator was mounted to the force gauge through a rubber band, and steady-state forces were measured when the rubber band was stretched due to actuation.

showed that the actuator force decreased with an increasing input frequency. This indicates that there is a significant damping effect from the dielectric fluid, suggesting that it is more efficient to drive our actuator at low synchronous speeds.

3. Sensor

Different methods have previously been used for position sensing of electrostatic actuators, including capacitive sensing with integrated separate electrodes (Niino et al., 1993), measurement of the input current driving the actuators (Zhang and Yamamoto, 2018), and transmission and reception of high frequency AC signals on top of the actuation AC input voltages (Nishijima et al., 2006). However, these approaches tend to either have a relatively high level noise or are only functional while the actuator is being driven.

We introduce a compact optical sensing method to measure the displacement of the proposed actuator with a high resolution. Our method utilizes the Moiré patterns which innately form on the actuator surface during motion due to the difference in skewing of the slider and the stator electrodes. This approach involving Moiré patterns for sensing has been widely used for precise measurement in the optics field (Tran et al., 1996; Wang et al., 2017; Yu et al., 2018). Since this non-invasive optical sensing system is uncoupled from the electrostatic actuation, the actuator position may be read continuously not only with negligible noise but also when the actuator is not driven.

Besides position sensing, the same optical system can be also used to measure the output force of the actuator. The force output of a synchronous actuator is related to its phase lag, which is also shown in our actuator model in Figure. 23 in the Appendix. With the high-precision position sensor, we were able to measure the position lag of the slider from the electrical input signal to estimate the instantaneous actuator force.

3.1. Moiré effect

In the slider and the stator films, the areas containing the metal electrodes have less luminous transmittance than the other areas consisting only of polymer. If both the slider and the stator have straight electrodes, the amount of light, transmitted from one surface to the other surface of the entire actuator in motion, changes in proportion to the overlapping areas of the electrodes of the slider and the stator. As a result, the surface of the actuator appears to alternate in brightness with flickering.

Since the stator electrodes in our actuator are skewed with V-shapes and the slider electrodes are straight, the electrodes of the two films are not parallel and overlap each other, creating a constant offset angle. Due to this acute angle between the slider and the stator electrodes, the overlap area changes along the longitudinal direction of the actuator, consequently changing the amount of transmitted light through the films. This phenomenon generates Moiré patterns on the actuator surface, as shown in Figure 7. The sequence of the figures on the bottom right corner of Figure 7 shows the movement directions of the patterns as the slider moves downward. Visually, the Moiré patterns appear to move in the direction perpendicular to the motion of the slider. Reversing the direction of the movement of the slider also reverses the movement direction of the patterns.

Figure 8 depicts the configuration of the electrodes of the slider and the stator when they have the same pitch, p. The overlapped electrodes display rhombuses of varying brightness. The angle difference between the electrodes, α, is dependent on the skew factor s_f and the width of the V-shape pattern of the electrodes. Lines intersecting the centers of the white rhombuses (represented by the dotted lines in Figure 8) allow light to penetrate the most, providing the brightest lines in the Moiré patterns on the actuator surface, since the slider and the stator electrodes are most overlapped along these lines. As the slider moves along the x-axis, the Moiré patterns move along the z-axis. Therefore, at a fixed location on the actuator, the bright lines are observed to pass periodically as the slider moves along in position.

Here, we define the distance between the lines intersecting the rhombus centers along the z-axis as the spatial period, s, which is the distance that the Moiré pattern travels along the z-axis when the slider travels one pitch along the x-axis. From the right triangle in Figure 8, the relationship between p, α, and s expressed as the ratio of the spatial period to the pitch is

$$s/p = \frac{1}{\sin\alpha} \tag{1}$$

By choosing an appropriate angle difference α, the spatial period s may be significantly amplified from the pitch p, providing the means for high-resolution position sensing. With the skew factor s_f set to be 1 in our actuator prototype design, we set the width of each diagonal (half of the V-shape) to

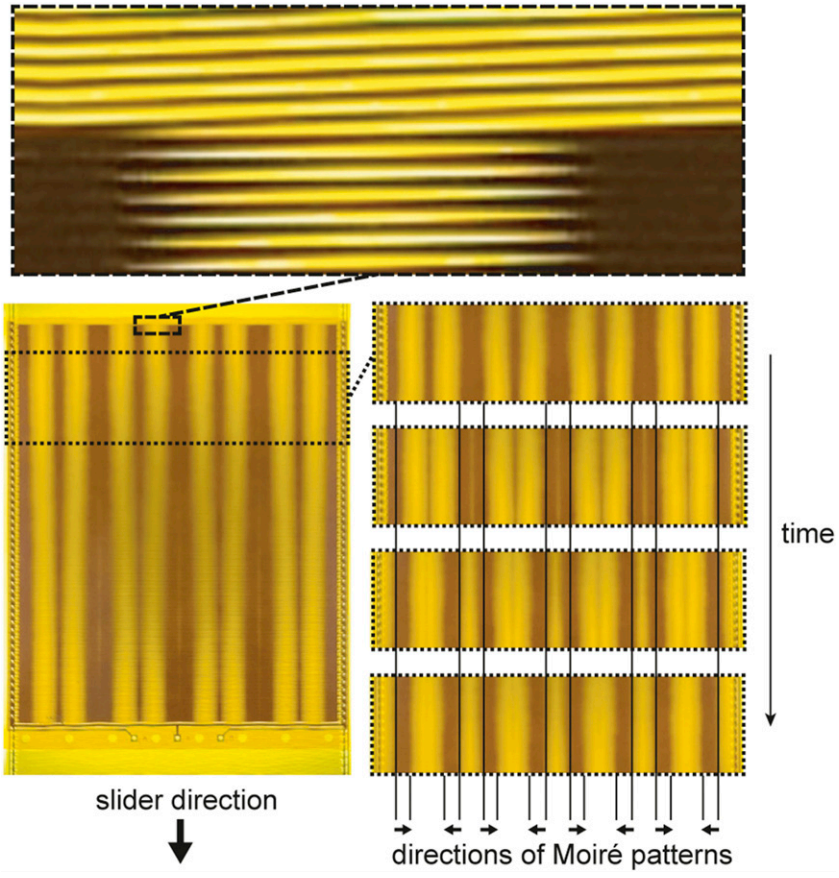


Figure 7. When the slider and the stator are overlapped, Moiré patterns are generated due to the angle difference between their electrodes. The movement direction of the patterns is perpendicular to the movement direction of the slider. A backlight was placed under the films to capture the patterns clearly.

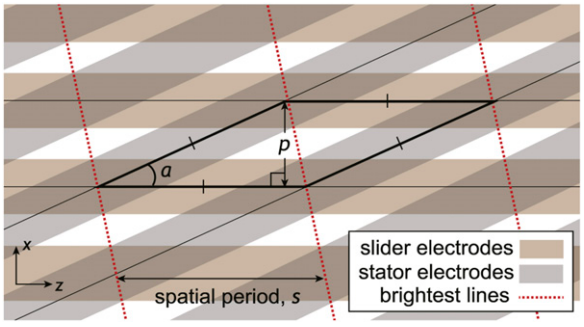


Figure 8. Configuration of the slider and stator electrodes. The lines intersecting the centers of rhombuses are brightest in the Moiré patterns.

9.25 mm, resulting in an angle difference of 1.86°. From (1), the spatial period s becomes 9.26 mm that closely matches the width of the diagonal in the electrode pattern. With such a design, a single Moiré pattern passes through a fixed position in our actuator prototype when the slider travels one pitch. If desired, however, it is possible to further amplify the sensing resolution by choosing a different design.

There are a total of eight Moiré patterns in Figure 7 since there are eight individual diagonal sections from the four V-shaped skew patterns in our actuator. When compared to the actual position of the slider, the resolution based on these Moiré patterns showed 30.85 times amplification (= s/p). The travel distance of the light patterns is then large enough to be easily and precisely measured by using millimeter-sized optical sensors.

3.2. Position sensing

We implemented a quadrature encoder utilizing the Moiré effect. By counting the number of patterns passing through and reading the movement direction of the pattern, we were able to measure the slider position in the same manner as incremental encoders in conventional robotic systems. Our quadrature encoder is composed of an optical sensing part and a light-emitting part, each consisting of two infrared phototransistors and two infrared light-emitting diodes (LEDs), respectively. The sensing part and the light-emitting part were attached to the outer surfaces of the actuator, opposing each other.

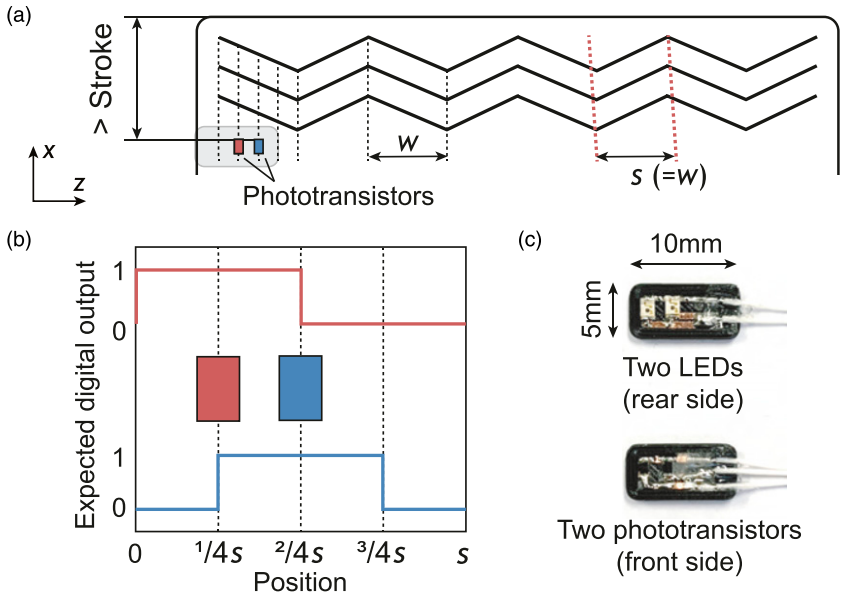


Figure 9. (a) Location of the sensing module. Two infrared LEDs and two infrared photodiodes are attached at the same location but on different sides. (b) Location of the two phototransistors inside the sensing module and their expected digital outputs. (c) Actual sensing module enclosed in cases that do not transmit external light.

Figure 9(a) shows the location at which the sensor modules were attached. While the module may be attached to any position along the *z*-axis, the two phototransistors should be located within the same section among the eight diagonal sections. In our implementation, the module was attached to read the outermost Moiré pattern. Along the *x*-axis, the module should be positioned inward from the top of the actuator in order to cover the full stroke length of the actuator. This ensures that the sensor always maintains sight of the Moiré patterns because the patterns occur only when the slider and the stator films overlap. The phototransistors were placed on the front side while the LEDs were placed on the rear side.

Figure 9(b) shows the placement of the two phototransistors and their expected digital outputs when a Moiré pattern passes through. The spacing between the phototransistors should be a quarter of the spatial period s to guarantee that positions are read at even intervals from the quadrature encoder. Figure 9(c) shows the actual encoder parts used in our prototype. The LEDs and the phototransistors were enclosed in rigid, light-impermeable cases to block any optical noise from the environment.

Figure 10 shows measured raw optical sensor signals and their converted digital values. Raw sensing signals were collected by pulling the slider at a speed of 1 mm/s using an external motorized test stand (ESM303, Mark-10), revealing a periodic response that every electrode pitch traveled. The shape of the signal, however, was decided by several factors, such as the electrode length ratio λ, the sensitivity of the phototransistor, and the intensity of the light source. To obtain equally-spaced digitized positions, we set the signal threshold according to the pulse width that provided a 50% duty cycle. After digitization, the incremental encoder of our prototype was able to read the actuator position with a resolution of 75 µm per tick.

Figure 11 shows one of the phototransistor signals and the instantaneous speed of the actuator when the actuator is driven. The instantaneous speed was measured through the optical tracking of the marker attached to the surface of the slider film by an optical camera filming at a sampling rate of 240 Hz, taken as the ground truth. During actuation, the raw sensor signals appeared to be distorted so that the pulses were unevenly distributed over time. This is because the actuator moved with an average speed that matched the synchronous speed but with an instantaneous speed that oscillated above and below the synchronous speed, as shown in Figure 11, also showing one of the two phototransistor signals. As observed from the figure, faster instantaneous speeds caused the Moiré patterns to occur at faster rates and resulted in shorter wavelengths in the raw sensor signal. Similarly, at slower speeds, the sensor signals appeared to have longer wavelengths. Nonetheless, on average, the phototransistor pulse rates matched the actuator's synchronous speed.

3.3. Force sensing

Actuation force was characterized by changing the weight of fluid in a cup in increments of 0.1 N measured using a digital scale (Scout Pro SP602, OHAUS). The load was attached to a string wound around a pulley and connected to the actuator. The phase lag was measured when the actuator was operated at the synchronous speed under the constant load. The results for frequencies of 1 Hz, 2 Hz, and 3 Hz with amplitudes of 1000 V, 1100 V, and 1200 V, respectively, are plotted in Figure 12, along

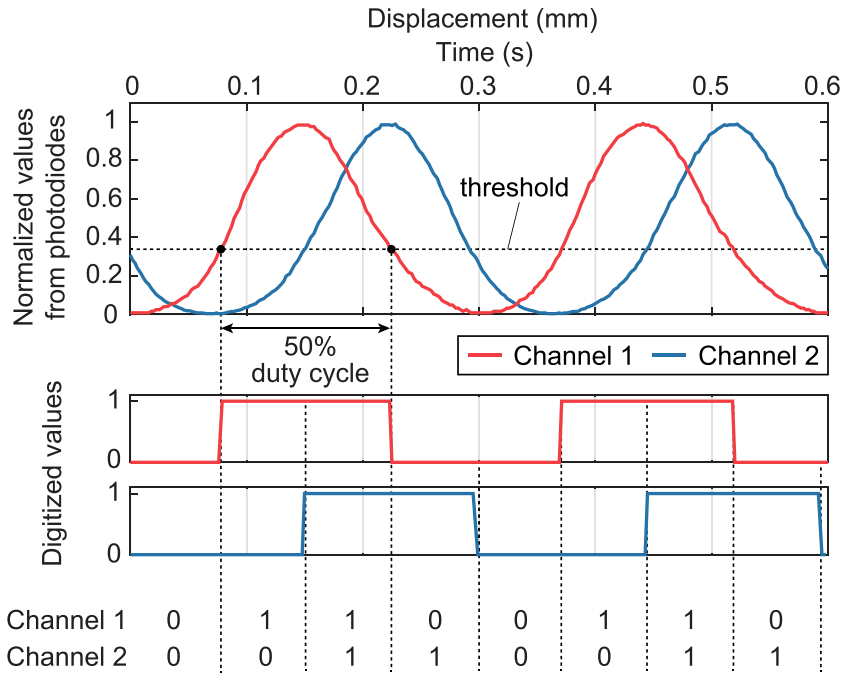


Figure 10. Raw sensing signals and calibration for even-interval position reading as a quadrature encoder. The slider was pulled by an external motorized stand at a constant speed of 1 mm/s. With the actuator electrode design, the position resolution is 75 µm.

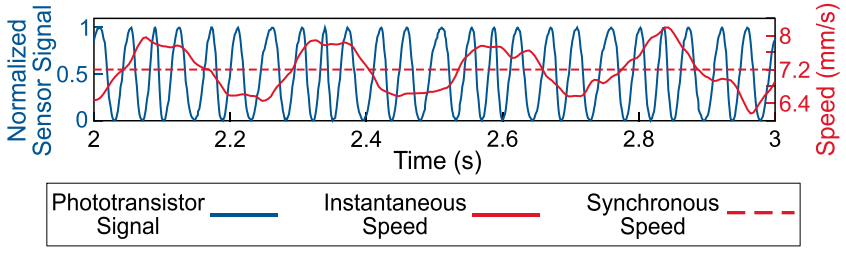


Figure 11. When the actuator is driven, the raw phototransistor signal appears to be distorted unevenly because of the varying instantaneous speed. Actuator is driven at 4 Hz and 1 kV. One of the two phototransistor signals with normalized amplitudes is plotted. The instantaneous speed is obtained via optical tracking of a marker.

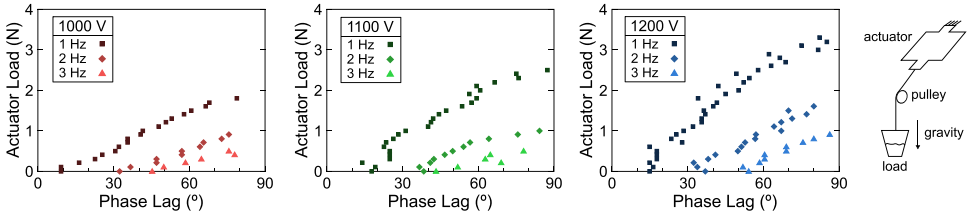


Figure 12. Experimental characterization of the actuator performance under load at frequencies of 1 Hz, 2 Hz, and 3 Hz for voltage amplitudes of 1000 V, 1100 V, and 1200 V in order of left to right. The experimental setup is depicted in the rightmost schematic where the load is a fluid weight inside a cup and connected to the actuator through a string on a pulley.

with a schematic of the experimental setup on the right. The phase lag was obtained by converting the time delay between the supplied electrical position and the position detected by the phototransistor, which was measured by comparing the waveforms of both using an oscilloscope (RTA4000, Rohde & Schwarz).

From the results, it is noted that increasing the voltage amplitude improved the actuator loading capacity, as expected from the model. Increasing the frequency, however, consistently reduced the loading capacity of the actuator. The phase lag with very light loads was also larger at higher speeds. In effect, the actuator's loading

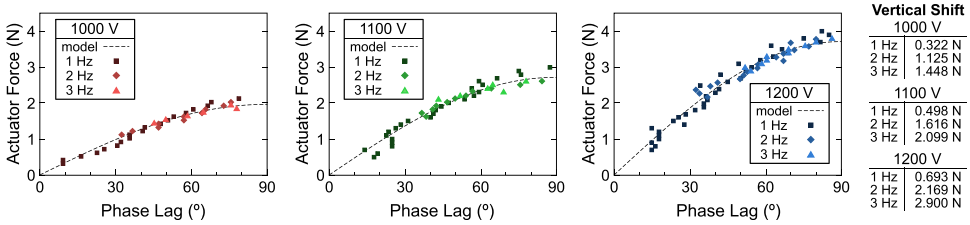


Figure 13. Experimentally obtained actuator loading results shifted along the *y*-axis by an amount that minimizes the sum of the vertical distances between the experimental data and the model force curve, the values of which are listed in the table on the right. Model curves are obtained from (24) and Figure 23 of the Appendix.

performance shows a trade-off with the operation speed, affecting not only the maximum stalling force but also the range of the allowable phase lag.

To better understand the effects of increasing the voltage amplitude and the frequency, Figure 13 shows the same experimental data obtained in Figure 12 in comparison with the force curve obtained by our model in Figure 23 in the Appendix, in this case, the experimental data were offsetted along the *y*-axis to mainly check the tendencies. The vertical offset values are provided in the table at the right of Figure 13. Considering that the experimental data fits the predicted force curves rather closely, the differences in the load performance of the actuator may be attributed to non-conservative forces that appear in the form of friction or viscosity.

In particular, comparison of the offsetted values shows that the non-conservative force increased not only as the frequency increased but also as the voltage increased. Increasing the voltage, while improving loading capacity, also increased the normal attraction force between the slider and the stator of the actuator, which in turn increased the frictional force. From the table in Figure 13, it can be seen that the frictional forces increased roughly in proportion to the supplied voltage amplitude. For the increases in frequency, the non-conservative forces increased in a similar manner because higher frequencies led to higher speeds that increased the viscous damping force. Damping forces are typically non-linear with respect to the speed, and the table values also increased in such a non-linear manner. The model force therefore predicts the actuator force under ideal circumstances without consideration of non-conservative forces, such as friction or damping that appear as vertical offsets in the force characterization plots.

Additionally, it is worth noting that our Moiré effect phototransistor sensor, although designed to sense the actuator position, can be used to estimate the actuator load through the measurement of the phase lag in comparison to the supplied electrical position. Once the phase lag is obtained in the sensors, the actuator load may be predicted using the model and the force characterization results by mapping the corresponding phase lag values to the actuator loads.

4. Brake

Our proposed actuator, while suitable for precise position control and velocity control, is in principle a synchronous actuator and is not ideal for consistent generation of a large static force. In robotic applications, however, holding and resisting of forces are often necessary to withstand static loads while maintaining a desired position. Accordingly, many robot systems are equipped with separate braking components in parallel with the actuators to effectively create static forces in a dissipative manner and also to protect the actuators from overheating when stalling. Similarly, there are several electrostatic brakes that have been recently investigated (Diller et al., 2018; Hinchet and Shea, 2020; Ramachandran et al., 2019; Zhang et al., 2019). Compared to existing rigid brake and clutch systems, these electrostatic brakes generate exceptionally high force densities with low power consumption while having thin and flexible forms.

This section introduces the combination of our actuator with an electrostatic brake. Such electrostatic brakes have been previously used for human augmentation (Diller et al., 2016; Ramachandran et al., 2019) or haptic applications Hinchet and Shea, 2020; Zhang et al., 2019) as they do not produce motions or active forces. However, in this work we combine the electrostatic brake with an active electromechanical component, that is, the electrostatic actuator, to reap the benefits of both. In other words, while the actuator creates the force and motion, the brake effectively holds the actuator position under a large external load. Both parts are easily integrated together as the two components share the same working principle, the power source, and the form factor. We also present and demonstrate two mode-switching strategies of our combined actuator based on the feedback from the optical sensor.

4.1. Design and characterization

The electrostatic brake consists of a slider film and a stator film, as shown in Figure 14(a). Similar to the actuator films, the brake slider film was fabricated with a copper electrode and polyimide dielectric films. The overall thicknesses of the slider and the dielectric film facing the stator for the brake are 100 µm and 13 µm, respectively. For the stator film, a 100 µm-thick conductive polymeric film (Velostat, 3M) was used with slits made along the longitudinal direction. These slits improve the engaging and releasing time of the brake by preventing lateral interferences (Diller et al., 2018). The slider and the

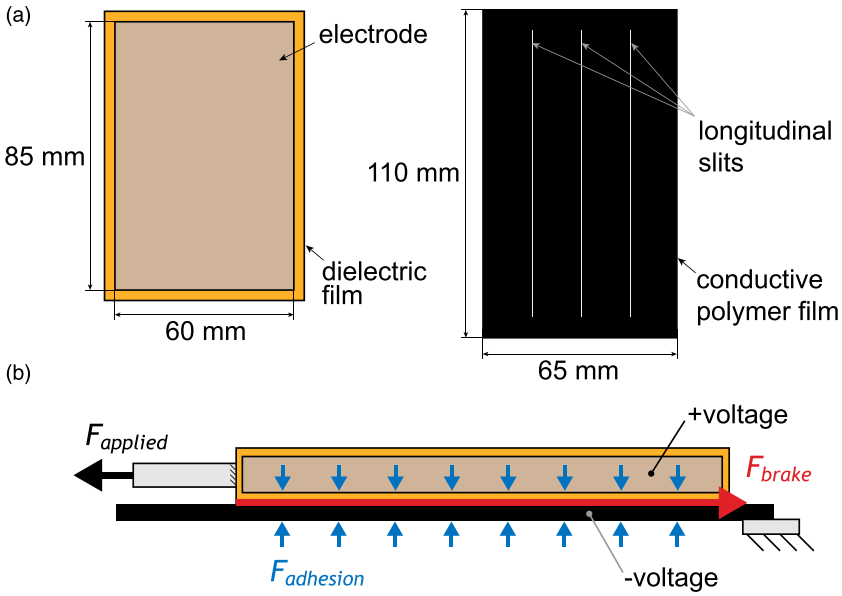


Figure 14. (a) Design of the brake slider and stator. (b) When there is a voltage difference between the brake slider and stator, friction occurs due to the electroadhesive force.

stator films of the brake were then attached to the frames, as shown in Figure 1(b).

Providing a voltage difference between the slider and the stator electrodes of the brake (see Figure 14(b)) generates an adhesive force F_{adhesion} of the two films, which then creates a frictional shear force F_{brake} such that

$$F_{\text{brake}} = \mu F_{\text{adhesion}} = \mu \frac{\epsilon_0 \epsilon_r A V^2}{2d^2} \tag{2}$$

where μ is the static friction coefficient between the slider and the stator films of the brake, ϵ_0 is the vacuum permittivity, ϵ_r is the relative permittivity of the dielectric film, V is the applied voltage, and d is the thickness of the dielectric film (Chen and Bergbreiter, 2017).

Figure 15 shows the maximum braking force of our brake according to differences in the applied voltage. For characterization, the brake was mounted on a motorized stand, attached to a force gauge, and pulled at a constant speed of 0.22 mm/s. The largest maximum braking force experimentally observed was 66 N at 1000 V where the brake showed a resistive force before disengaging.

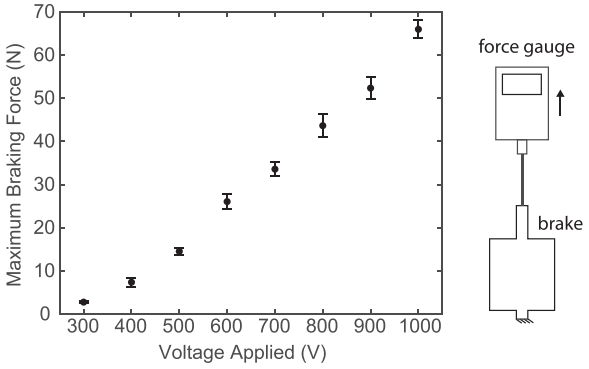


Figure 15. Averaged maximum braking force of the electroadhesive brake. Each condition was measured five times. Each error bar indicates one standard deviation of the mean.

4.2. Mode switching and one-DOF manipulator

The proposed actuator package is readily applicable to robot systems as a thin and compact alternative to conventional actuators. As an example, we assembled the actuator system with a one-DOF robot arm manipulator for testing simple weight-lifting, one of the most basic motions that often occur during physical interactions with humans. Through this experiment we demonstrate the efficacy of mode switching between the actuator and the brake.

Figure 16(b) shows the demonstration setup with the one-DOF manipulator. The manipulator was positioned perpendicular to the ground with our actuator package mounted to mimic the biceps of the human arm. Contraction forces of the actuator were transmitted to the pulley at the joint that lifted the forearm. The pulley radius was 25 mm, and the moment arm from the joint to the end-effector was 200 mm. A transparent plastic cup hung by a string was placed at the end of the forearm to act as a weight containing fluid. When a load was applied to the end-effector, the actuator received eight times of the load. The weight of the forearm and the cup was counterbalanced in advance using an additional weight in the opposite direction of the joint to compensate for the gravity when unloaded.

Since the position sensor measures the relative displacement from the first measured position using the

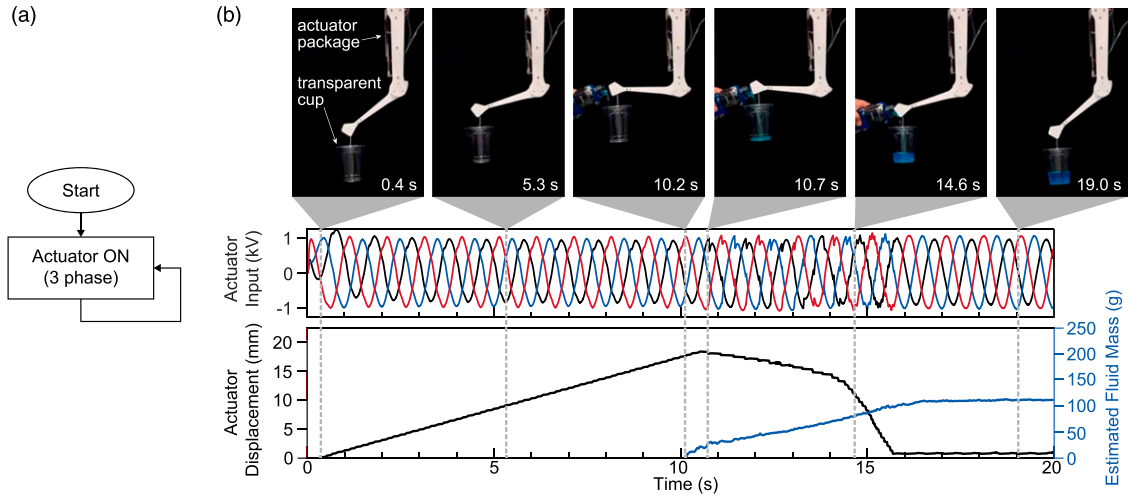


Figure 16. Manipulator demonstration only with the actuator. (a) Control flow. (b) Experimental results.

incremental encoder method, a calibration process was performed by driving the actuator to its maximum extension and setting the position to zero to measure the absolute displacement.

The experimental procedure started with the manipulator lifting the plastic cup by driving the actuator. When the cup rose to a certain level of height, fluid is poured into the cup until the weight of the fluid is greater than the force applied by the actuator, upon which different closed-loop schemes were chosen to handle the excessive load.

4.2.1. Actuator alone. We first implemented the one-DOF manipulator only with the actuator—without using the brake—as the control flow in Figure 16(a) indicates. The voltage amplitude and the frequency of our three-phase actuator input were 1000 V and 1 Hz, respectively. Figure 16(b) shows the results. The actuator position was read using the integrated optical sensor. The mass of the fluid was estimated based on the filled area of the cup shown in the video using a motion-graphics software tool (Blender, Blender Foundation), assuming that the cup is in the shape of a truncated cone. During the first half of the entire period, the manipulator stably lifted the cup at the synchronous speed. When the fluid mass increased beyond approximately 30 *g*, the load became greater than the pull-out force of the actuator, and the two films of the actuator began to slide backwards while the actuator was still driven. A noticeable noise was generated at the actuator input when backsliding occurred. After the forearm was fully lowered by the load, the actuator did not create any motions and did not receive any energy from backsliding, resulting in distortion-free sine waves at the actuator input signals.

4.2.2. Closed-loop using actuator position. We then implemented activation of the electrostatic brake by detecting the actuator position. Figure 17(a) shows the control flow. While driving the actuator, the system reads the actuator position and checks for any backsliding through the encoder with a threshold of 1 mm. Upon detecting backslip in position greater than the threshold, the system turns off the actuator and activates the brake. The result in Figure 17(b) shows that the brake maintained the forearm position while the cup was filled with the fluid up to approximately 190 *g*, a much greater load than the actuator pull-out force. However, there is an innate drawback in this control approach. There must exist some unavoidable backsliding in the actuator position in order to activate the brake. While the threshold for detecting backsliding may be set to be a small value to minimize the amount of initial backsliding, it is unavoidable to have a non-zero value so that the system detects whether the load has exceeded actuator's pull-out force or not.

4.2.3. Closed-loop using actuator phase lag. To overcome the drawback of the position-based control loop, we demonstrated a control flow that activates the brake through force estimation of the actuator. As shown in Figure 18(a), the system calculated the actuator phase lag by comparing the actuator position signal from the optical sensor and the actuator input signal. For practical implementation, the force estimation technique referenced the peaks of the sensor and the input signals for simplicity. Although the actuator may be driven fully to the pull-out force at a phase lag of 90°, we set the threshold for activating the brake conservatively to an actuator phase lag of 30°. The demonstration results are presented in Figure 18(b), which shows that the brake was activated when the phase lag reached 50° and was able to switch to the holding mode with no backsliding, maintaining its position while holding a fluid mass of 145 *g*. While the low sampling rate of the force estimation may be improved by performing careful real-time calculations of the phase lag, we were still able to demonstrate the mode switching of our actuator system without any backsliding by using the force estimation feedback.

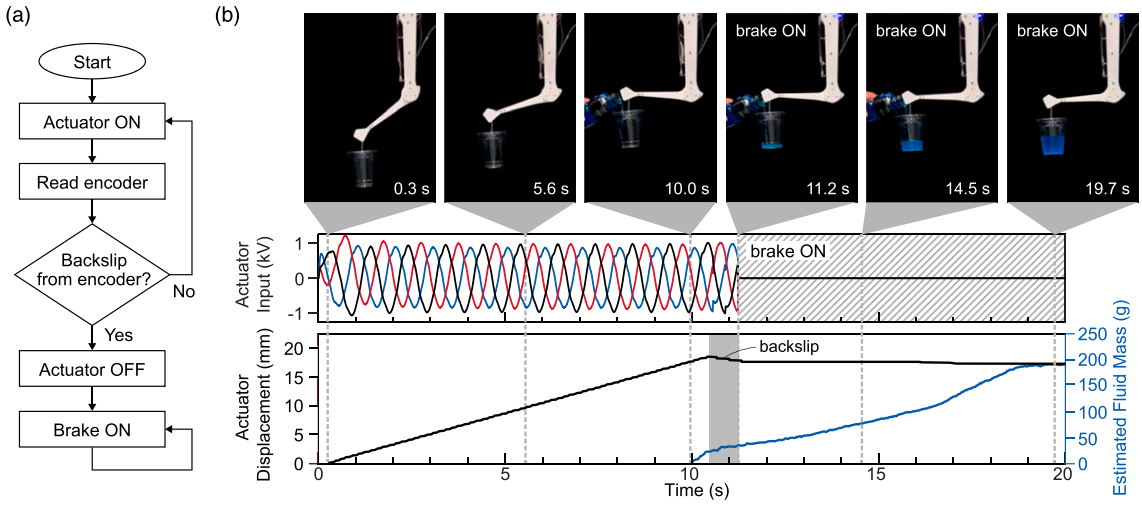


Figure 17. Manipulator demonstration with brake activation using actuator position sensing. (a) Control flow. (b) Experimental results.

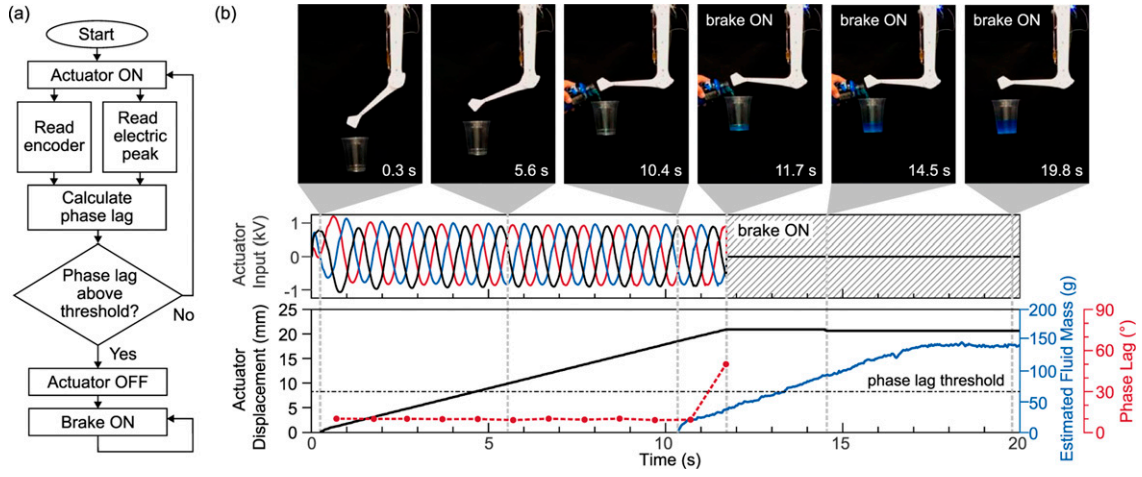


Figure 18. Manipulator demonstration with brake activation using actuator force estimation. (a) Control flow. (b) Experimental results.

5. Discussion

This paper proposed several design aspects of linear electrostatic actuator systems for practical implementation.

First, the electrostatic actuator films were packaged into a compact sealed enclosure to contain a dielectric fluid for improved performance that also played a role of guide rails for linear motions. The thin and flexible form of the actuator package allowed it to be easily integrated with various robotic systems with limited design space, for example, by inserting the actuator package into a narrow gap or attaching it to a curved surface. Although we demonstrated the actuator with a single prototype with set dimensions for the stator and the slider films, in principle the design is not constrained by these dimensions and may be adjusted to provide wide ranges of the linear stroke length and the actuation forces; the greater difference between the lengths of the stator and the slider films creates the longer stroke length, and the larger area of the slider film generates the larger force. In addition, the small thickness of the actuator may be utilized to further increase the actuation force by stacking multiple electrode film layers or actuator packages in parallel without significantly changing the overall thickness and weight, as discussed in previous work (Niino et al., 1994).

Second, the actuator has a function of proprioceptive feedback using optical sensing of the Moiré patterns innately generated on the actuator surface due to the difference in the skew angles of the stator and the slider electrodes. Based on the design of the electrodes, the Moiré patterns may significantly amplify the displacement of the actuator, providing a high resolution for position sensing. In our prototype, we were able to reliably read increments of 75 μm although detection of even smaller increments was possible with further amplification. The same optical sensor was also used to measure the actuator force based on the characteristic of synchronous operation whereby the force was related to the phase lag. This sensing principle does not

require modification of the sensor size for different actuator sizes but only requires detection of the actuator input and the calculation of the delay between the actuator input and the sensor output. Therefore, the sensing part remains compact even though the size of the actuator is increased.

Finally, the electrostatic actuator was integrated with an electrostatic brake. The brake is also thin and flexible, and does not significantly change the overall form factor of the actuator after integration. Demonstration with a one-DOF robot arm manipulator showed that the combination of the brake and the actuator was able to not only lift a weight at the end-effector but also hold the load in a fixed position—a task that cannot be accomplished with only one of the two components. From the results, we showed that our combined actuator was applicable to other robot systems that need to generate continuous motions with a load and maintain their physical configurations under the load.

In spite of the current advantages of the proposed system, there is a room for further improvement. The selection of the materials used for the actuator and the brake films in our prototype were limited by their accessibility in the market industry. However, the optimum design for the best performance will be possible with a wide range of available materials. For example, previous research demonstrated an improved performance of an electrostatic clutch by using poly (vinylidene fluoride-trifluoroethylene-chlorotrifluoroethylene) (*p*(VDF-TrFE-CTFE)) as a dielectric material due to its high dielectric constant (Hinchet and Shea, 2020). Fabricating the actuator and the brake films with a material with a higher dielectric constant than polyimide will also increase the maximum forces of the actuator and the brake. This implies that such films generate the same amount of force at a lower input voltage, reducing potentially hazardous sparks or short circuits. Moreover, although our actuator prototype was mostly made of a thin and flexible material, the optical sensor that we used did not comply with the characteristic. It is the thickest and the only rigid part in our actuator package. By replacing the sensor with a thin and flexible component, the flexibility and the compact form factor of the actuator package may be fully utilized. Optical fibers can be an option to detect the Moiré patterns as an alternative to the pairs of phototransistors and LEDs, enabling seamless integration of the actuator package with various robotic systems.

While we demonstrated the application of the proposed actuator with a traditional robotic arm in this paper, the advantages of the actuator allow it to be utilized in more diverse applications. Namely, the actuator is lightweight, compliant, and also compact. For applications, such as wearable robots and devices, the lightweightness helps reduce burdens on the wearer, and the compliance allows the actuator to better conform to the surfaces of the body of the wearer. Other applications, such as those for outer space also benefit from lightweight actuators with the increased fuel efficiency, and any strict space constraints can be readily satisfied with the compact form factor.

6. Conclusion

In this work, we presented a concept of an actuator system consisting an electrostatic actuator, an optical sensor, and an electrostatic brake. We derived electrostatic and electromechanical models to design the actuator electrodes. With the introduction of an oil-embedded sealing package, we conducted experiments with various amplitudes and frequencies of the voltage inputs. The measured actuation force showed a high agreement with the model predictions at low synchronous speeds. Then, we integrated an optical sensor into the actuator to read the Moiré patterns on the actuator surface. The digitized signals from the sensor were used for noise-free, high-resolution position sensing, and the raw signals were used to calculate the phase lag for estimating the actuation force. For the actuator and the brake combined together in parallel, we introduced two mode-switching strategies with closed-loop control using the optical sensor—with and without positional backsliding. The efficacy of the proposed system was demonstrated with a one-DOF manipulator, validating the use of the parallel structure and the mode switching in a load-lifting and holding task. The overall actuator system is compact and lightweight even with the sensor and the brake components combined.

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Declaration of conflicting interests

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Appendix

Actuator modeling and design

Here, we derive the output force model of our electrostatic actuator. Compared to previous work (Yamamoto et al., 2006), our model has an explicit form of a capacitance matrix with the assumption of simplified electric fields between the electrodes, and therefore, it is not necessary to measure the capacitance between the electrodes experimentally in order to calculate the output force. Based on the model, we determine the dimensions of the electrodes, aiming to generate the output force with a large magnitude and a small ripple.

Electrostatic model

Figure 19 shows the model of our electrostatic actuator. The slider and the stator were patterned with electrodes of the same thickness t, length L, and spacing, or gap g. The electrodes were encapsulated in an insulating film with a dielectric constant ϵ_f, and lubricant with a dielectric constant ϵ_l was used to ease the relative motion between the slider and the stator along the x-axis. The vertical distance between the electrodes of the slider and the stator is the sum of the lubrication thickness t_l and the unilateral thicknesses $t_{f,sl}$ and $t_{f,st}$ of the insulating film for the slider and the stator, respectively.

Assuming the electrode length is much larger than the electrode thickness ($L \gg t$), the fringing effects of the electric fields from the electrodes become negligible. The model therefore assumes that the electric fields related to a specific charge q are all uniform and exist only in a region that connects the length-wise dimension of the electrodes in straight lines, as depicted by the dashed lines in Figure 19. The electric fields are notated as $\boldsymbol{E}_{a,b}$ for a field going from charge q_a to q_b. An additional subscript f or l is added to distinguish the field inside the insulating film or the lubricant, respectively. The directions for the electric fields are chosen arbitrarily but kept consistent throughout the derivation of the charge relations. For a given electrode, only the fields from the nearest five electrodes are considered in the model. Fields from charges further away are considered negligible and are excluded.

From the integral forms of Maxwell's equations for quasi-static electric field systems,

$$\oint_S \varepsilon \boldsymbol{E} \cdot \boldsymbol{n} \mathrm{d}A = q \tag{3}$$

and

$$\oint_C \boldsymbol{E} \cdot \mathrm{d}\boldsymbol{l} = 0 \tag{4}$$

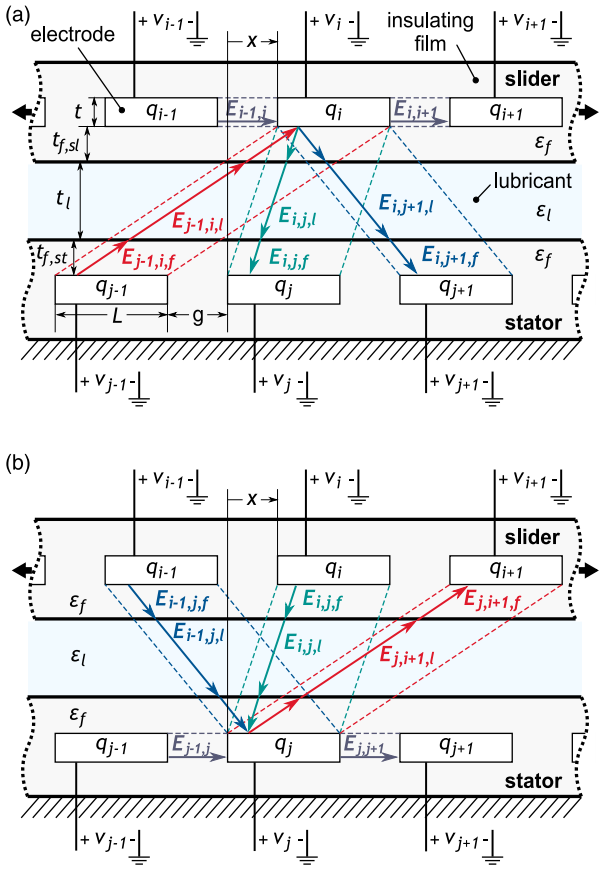


Figure 19. Cross-sectional schematic of an electrostatic actuator with repeatedly patterned electrodes in a slider and stator configuration. Electric fields are modeled as uniform fields that exist only within the regions indicated by dashed lines and are drawn for the ith slider electrode with charge q_i (top) and for the jth stator electrode with charge q_j (bottom). Charges that are further away are not shown in the schematic; their electric fields are considered negligible and are excluded from the model.

where (3) provides the charge relations for q_i and q_j with respect to the electric fields, and (4) relates the electric fields to the terminal voltages v_i and v_j supplied to the ith electrode in the slider and the jth electrode in the stator, respectively. Substituting the electric fields with the terminal voltages of nearby electrodes $\boldsymbol{v}_i = (v_{i-1}, v_i, v_{i+1})^{\mathrm{T}}$ and $\boldsymbol{v}_j = (v_{j-1}, v_j, v_{j+1})^{\mathrm{T}}$ yields

$$q_i = \boldsymbol{c}_{\mathrm{s}}\boldsymbol{v}_i - \boldsymbol{c}_{\mathrm{m}}\boldsymbol{v}_j \tag{5}$$

and

$$q_j = -\boldsymbol{c}_{\mathrm{m}}\mathbf{J}_3\boldsymbol{v}_i + \boldsymbol{c}_{\mathrm{s}}\boldsymbol{v}_j \tag{6}$$

where $\mathbf{J}_3 \in \mathbb{R}^{3\times 3}$ is the exchange matrix (unity along the anti-diagonal and zero otherwise) for self-capacitance $\boldsymbol{c}_{\mathrm{s}} = (-C_0, C_\Sigma, -C_0)$ and mutual capacitance $\boldsymbol{c}_{\mathrm{m}} = (C_{x+p}, C_x, C_{x-p})$ such that

$$\begin{aligned} C_0 &= \frac{\varepsilon_{\mathrm{f}} t}{g} \\ C_x &= C_{\text{series}}\cos^2(\theta_x) \\ C_{x-p} &= C_{\text{series}}\cos^2(\theta_{x-p}) \\ C_{x+p} &= C_{\text{series}}\cos^2(\theta_{x+p}) \\ C_\Sigma &= 2C_0 + C_x + C_{x-p} + C_{x+p} \end{aligned} \tag{7}$$

where

$$\frac{1}{C_{\text{series}}} = \frac{t_{\mathrm{f,sl}}}{\varepsilon_{\mathrm{f}} L} + \frac{t_{\mathrm{l}}}{\varepsilon_{\mathrm{l}} L} + \frac{t_{\mathrm{f,st}}}{\varepsilon_{\mathrm{f}} L} \tag{8}$$

results from the equivalent series capacitance relation of parallel plate capacitors. This equivalent capacitance C_{series} has a cosine-squared relation with the angles between the respective electric field vector and the normal of the length-wise surface of the electrode. From trigonometric identities, $\cos^2\theta = 1/2 + \cos(2\theta)/2$. Yamamoto et al. previously approximated measured capacitances using sine functions up to third-order harmonics (Yamamoto et al., 2006); our model is similar in that it considers the zeroth and second-order harmonic.

Evaluation of the cosine-squared terms gives

$$\begin{aligned} \cos^2(\theta_x) &= \frac{d^2}{d^2 + x^2} \\ \cos^2(\theta_{x-p}) &= \frac{d^2}{d^2 + (x-p)^2} \\ \cos^2(\theta_{x+p}) &= \frac{d^2}{d^2 + (x+p)^2} \end{aligned} \tag{9}$$

where $d = t_{\mathrm{f,sl}} + t_{\mathrm{l}} + t_{\mathrm{f,st}}$ is the vertical distance between the electrodes in the slider and the stator, and $p = L + g$ is the electrode pitch.

For n distinct voltages in the slider and the stator that are supplied to the electrodes in a repeating sequential order (i.e., $v_{i+n} = v_i$ and $v_{j+n} = v_j$), the complete charge relations are obtained by writing (5) and (6) for $i = 1, \ldots, n$ and $j = 1, \ldots, n$. The resulting capacitance matrix $\mathbf{C}$ is symmetric and in the form of

$$\mathbf{C} = \begin{bmatrix} \mathbf{C}_{\mathrm{s}} & -\mathbf{C}_{\mathrm{m}} \\ -\mathbf{C}_{\mathrm{m}}{}^{\mathrm{T}} & \mathbf{C}_{\mathrm{s}} \end{bmatrix} \tag{10}$$

where the self-capacitance matrix $\mathbf{C}_{\mathrm{s}} \in \mathbb{R}^{n\times n}$ and the mutual capacitance matrix $\mathbf{C}_{\mathrm{m}} \in \mathbb{R}^{n\times n}$ are both circulant matrices with $\boldsymbol{c}_{\mathrm{s}}$ and $\boldsymbol{c}_{\mathrm{m}}$ centered along the diagonal, respectively. The former matrix relates slider charges $\boldsymbol{q}_{\mathrm{sl}} = (q_{i=1}, \ldots, q_{i=n})^{\mathrm{T}}$ to slider voltages $\boldsymbol{v}_{\mathrm{sl}} = (v_{i=1}, \ldots, v_{i=n})^{\mathrm{T}}$ and stator charges $\boldsymbol{q}_{\mathrm{st}} = (q_{j=1}, \ldots, q_{j=n})^{\mathrm{T}}$ to stator voltages $\boldsymbol{v}_{\mathrm{sl}} = (v_{j=1}, \ldots, v_{j=n})^{\mathrm{T}}$; the latter matrix relates slider or stator charges to voltages of the other.

Electromechanical model

From a viewpoint of electromechanics, our electrostatic actuator is mechanical system coupled with an electric field that converts electrical energy into mechanical work. Assuming that there is no loss during conversion, all supplied electrical energy W_{e} should be equal to the output, mechanical work W_{m}. When differentiated with respect to the thrust direction x, W_{m} provides the actuation force f such that

$$f = -\frac{\partial W_{\mathrm{m}}}{\partial x} = -\frac{\partial W_{\mathrm{e}}}{\partial x} \tag{11}$$

For quasi-static electromagnetic conditions with no changing magnetic fields, the electrical energy for an electric field system is

$$W_{\mathrm{e}} = \int v\mathrm{d}q \tag{12}$$

where q is the charge and v is the electrical potential. In a linear system containing multiple charges $\boldsymbol{q}$ each with different electrical potentials $\boldsymbol{v}$,

$$\boldsymbol{q} = \mathbf{C}\boldsymbol{v} \tag{13}$$

where the matrix $\mathbf{C}$ represents the capacitance of the system. This linear relation allows the calculation of the actuation force directly from coenergy W_{e}':

$$f = \frac{\partial W_{\mathrm{e}}'}{\partial x} = \frac{\partial}{\partial x}\int \boldsymbol{q}\mathrm{d}\,\boldsymbol{v} = \frac{1}{2}\boldsymbol{v}^{\mathrm{T}}\frac{\partial \mathbf{C}}{\partial x}\boldsymbol{v} \tag{14}$$

Taking the derivative of (10) with respect to the x-axis,

$$\frac{\partial \mathbf{C}}{\partial x} = \begin{bmatrix} \frac{\partial}{\partial x}C_\Sigma\mathbf{I} & -\frac{\partial}{\partial x}\mathbf{C}_{\mathrm{m}} \\ -\frac{\partial}{\partial x}\mathbf{C}_{\mathrm{m}}{}^{\mathrm{T}} & \frac{\partial}{\partial x}C_\Sigma\mathbf{I} \end{bmatrix} \tag{15}$$

where $\mathbf{I} \in \mathbb{R}^{n\times n}$ is the identity matrix. The disappearance of the off-diagonal terms in $\mathbf{C}_{\mathrm{s}}$ confirms that the slider-to-slider and stator-to-stator self-capacitances in our model do not contribute to actuator force.

From (14), the actuator force along the x-axis is then

$$f = \frac{1}{2}\begin{bmatrix} \boldsymbol{v}_{\mathrm{sl}}{}^{\mathrm{T}} & \boldsymbol{v}_{\mathrm{st}}{}^{\mathrm{T}} \end{bmatrix} \frac{\partial \mathbf{C}}{\partial x} \begin{bmatrix} \boldsymbol{v}_{\mathrm{sl}} \\ \boldsymbol{v}_{\mathrm{st}} \end{bmatrix} \tag{16}$$

which is simplified and rearranged into the sum of the forces

$$f = \sum_{i=1}^{n} f_{i,j} \tag{17}$$

where $f_{i,j}$ is the component force generated by the voltage differences between the ith charge in the slider and the nearby $(j-1)^{\mathrm{th}}$, j^{th}, and $(j+1)^{\mathrm{th}}$ charges in the stator such that

$$f_{i,j} = \frac{1}{2} C_{\mathrm{series}} \begin{pmatrix} (v_i - v_{j-1})^2 & \frac{\partial}{\partial x}\cos^2(\theta_{x+p}) \\ +(v_i - v_j)^2 & \frac{\partial}{\partial x}\cos^2(\theta_x) \\ +(v_i - v_{j+1})^2 & \frac{\partial}{\partial x}\cos^2(\theta_{x-p}) \end{pmatrix} \tag{18}$$

Supplying sinusoidal, three-phase voltages ($n = 3$) in single-frequency operation, as explained in Niino et al. (1993), allows the voltages in the slider v_i and the stator v_j to be expressed as

$$v_i = V \sin\left(\theta + \frac{2\pi}{3} i\right) \tag{19}$$

$$v_j = V \sin\left(\theta - \frac{2\pi}{3} j\right) \tag{20}$$

where V is the voltage amplitude and θ is the electrical angle of the supplied voltage.

As the slider moves along the stator, the ith electrode in the slider aligns with many different electrodes in the stator; the index of the aligned stator electrode j may be expressed as the kth offset from the ith slider electrode as $j = i + k$, where $k = \lfloor x/p + 1/2 \rfloor$ such that $\lfloor\,\rfloor$ denotes the floor function.

Applying (19) and (20) to (18) and summing over all the distinct voltages ($i = 1, \ldots, n$) then provides

$$\sum_{i=1}^{n=3} f_{i,i+k} = f_k \tag{21}$$

where

$$f_k = \frac{1}{2} C_{\mathrm{series}} V^2 \times$$

$$3\begin{pmatrix} \left(1 + \cos\left(2\theta - \frac{2\pi}{3}(k-1)\right)\right) & \frac{\partial}{\partial x}\cos^2(\theta_{x+p}) \\ +\left(1 + \cos\left(2\theta - \frac{2\pi}{3}k\right)\right) & \frac{\partial}{\partial x}\cos^2(\theta_x) \\ +\left(1 + \cos\left(2\theta - \frac{2\pi}{3}(k+1)\right)\right) & \frac{\partial}{\partial x}\cos^2(\theta_{x-p}) \end{pmatrix} \tag{22}$$

is the cross-sectional actuator force for a constant k offset and is defined only for $x \in [-\mathrm{p}/2, \mathrm{p}/2)$. For the force at an arbitrary actuator position x_{pos}, the periodicity of the model is utilized for calculation of x from $x = x_{\mathrm{pos}} - kp$.

Model-based Electrode Design

Since our actuator is synchronous, the actuation force is a function of phase lag δ defined as the difference between the angular position of the apparent traveling electrical potential wave and the actuator position, that is, $\delta = \phi_e - \phi_x$, where ϕ_e is twice the electrical angle of the supplied voltage ($\phi_e = 2\theta$) and ϕ_x is the angular position of the actuator ($\phi_x = (2\pi/3p)x_{\mathrm{pos}}$).

By introducing the electrode length ratio $\lambda = L/p$, the electrode length and gap become $L = \lambda p$ and $g = (1-\lambda)p$, respectively. For straight electrode patterns (i.e., no skewing) of width w along the z-axis (defined as perpendicular to the actuator cross-section), the actuator force simply becomes the electrode width w multiplied by the cross-sectional actuator force f_k. Normalizing the electrode pitch (as well as other parameters V, w, C_{series}, and d) to unity allows the actuator force to be plotted for different electrode ratios λ, shown in Figure 20. Force curves have been averaged over all the electrical angles θ. It is clear from Figure 20 that the larger values for the electrode length ratio λ provide the larger actuator forces. In practice, relatively high voltages are however supplied to the actuator for generation of force at useful magnitudes, making use of large electrode length ratios λ impractical due to the risk of dielectric breakdown between neighboring electrodes at small electrode gaps g. Therefore, we chose $\lambda = 4/6$ (= 2/3) for the electrode length ratio in our actuator design to prevent a short circuit and dielectric breakdown.

Instead of straight electrode patterns with width w, skewing is introduced in the electrode design to help

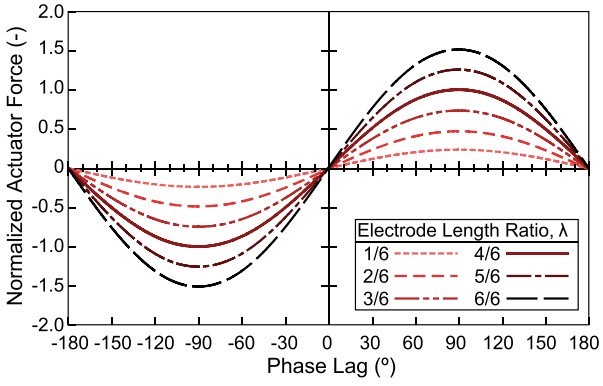


Figure 20. Actuator force for different electrode length ratios λ in the model (without skewing). Supplied voltage amplitude V, electrode pitch p and width w, capacitance C_{series}, and vertical distance d are all normalized to unity. The force is averaged over all electrical angles θ.

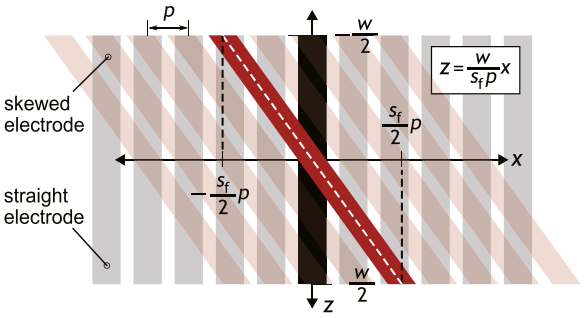


Figure 21. Generic linear electrode skew pattern with width w along the z-axis. Skew factor s_f is defined as the ratio of the skewing length along the x-axis to the electrode pitch p.

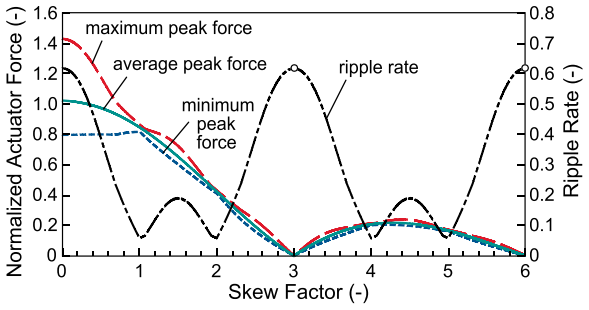


Figure 22. Minimum, average, and maximum actuator force and the ripple rate as a function of the skew factor s_f with supplied voltage amplitude V, electrode pitch p and width w, capacitance C_{series}, and vertical distance d are all normalized to unity.

reduction of force ripple of the actuator (Yamamoto et al., 1998). For a generic linear skewing pattern with width w and skew factor s_f, the electrode shape along the z-axis is expressed as $z = wx/(s_f p)$, as illustrated in Figure 21. With skewing, the actuator force is obtained from the integration of the cross-sectional actuator force along the z-axis:

$$f_{\text{skew}} = \int_{-w/2}^{w/2} f_k \mathrm{d}z = \frac{w}{s_f p} \int_{-s_f p/2+x_{\text{pos}}}^{s_f p/2+x_{\text{pos}}} f_k \mathrm{d}x \tag{23}$$

for which the evaluation of the integral requires splitting the limits into regions where $k = \lfloor x/p + 1/2 \rfloor$ is constant. For such a region, the force is

$$f_{\text{skew, constant } k} = \frac{1}{2} \frac{w}{s_f p} C_{\text{series}} V^2 \times$$
$$3 \begin{pmatrix} (1 + \cos(2\theta - 2\pi(k-1)/3))\cos^2(\theta_{x+p}) \\ +(1 + \cos(2\theta - 2\pi k/3))\cos^2(\theta_x) \\ +(1 + \cos(2\theta - 2\pi(k+1)/3))\cos^2(\theta_{x-p}) \end{pmatrix}_{x_0 - kp}^{x_f - kp} \tag{24}$$

where x_f and x_0 are the appropriate upper and lower limits for the region, respectively.

The rate of force ripple is then defined in the same way as in Yamamoto et al. (1998):

$$\text{ripple rate} = \frac{\text{maximum } f_{\text{peak}} - \text{minimum } f_{\text{peak}}}{\text{average } f_{\text{peak}}} \tag{25}$$

where f_{peak} is the peak (maximum) actuator force across all electrical angles ϕ_e for a fixed actuator position x_{pos}. The maximum, the minimum, and the average f_{peak} values are obtained by considering all actuator positions. Figure 22 shows the plots of the forces and the ripple rate as a function of skew factor s_f for our model with $\lambda = 2/3$ and other parameters normalized to unity as before. The ripple rate has a period of three beginning at its maximum value and is

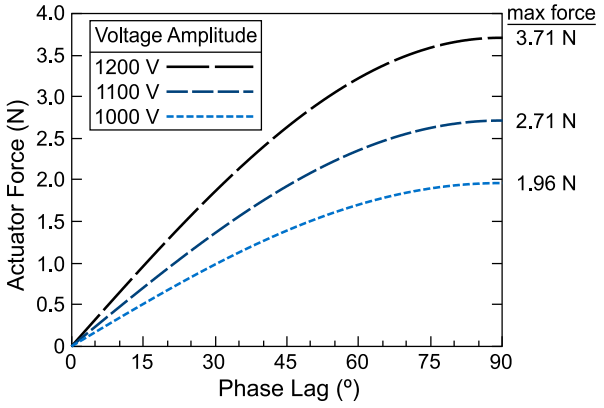


Figure 23. Predicted actuator prototype force from our model (s_f = 1 and $\lambda = 2/3$), averaged over all electrical angles θ. The force is multiplied for every $n = 3$ repeating electrodes, for the eight repeating skewed segments, and for the sandwich of the slider between two stators.

symmetric for every 1.5, showing large ripple rates for the range $s_f \in [1, 2]$. While the results for this range differ from the results previously reported by Yamamoto et al. (1998) whereby the ripple rate continues to decrease, both models agree that the ripple rate is largest near $s_f = 0$ and $s_f = 3$ and is lower in-between. They also agree that the average force decreases monotonically within the first period where . For our actuator prototype design, we therefore used a skew factor of $s_f = 1$ to alleviate the force ripple without over-compromising the average force.

Figure 23 is the actuator force predicted by our model for our prototype design ($s_f = 1$ and $\lambda = 2/3$). The force is averaged over all electrical angles θ and multiplied accordingly to account for the total number of electrodes, the eight skewed segments, and the sandwiched layers (details of which are described in the following section). The estimated maximum pull-out forces occur at 90° and are 1.96 N, 2.71 N, and 3.71 N for supplied voltage amplitudes of 1000 V, 1100 V, and 1200 V, respectively.